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Data Mining and Machine Learning Project

Musical Genre Classifier

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1 Introduction

Music represents a rapidly growing multi-billion-dollar market, deeply present in everyday life. People listen to music in many ways, often without even realizing it, whether it's the background tunes in a store, advertisements on the radio, or the soundtrack of a movie. Music also has a profound impact on human behavior and emotions, influencing thoughts and moods through rhythm, melody, and tempo.

Today, creating music has become easier than ever, thanks to increasingly accessible software tools, while digital platforms such as Spotify and iTunes make distribution effortless. As a result, these platforms manage an immense number of audio tracks and strive to provide personalized experiences that match individual user preferences.

Classifying songs by genre is a crucial step in enabling these personalized experiences. This project explores and proposes a method for classifying the genre of audio tracks by extracting easily obtainable spectral and musical features. The approach leverages widely used, interpretable Machine Learning models, focusing not only on accuracy but also on efficiency, with techniques for feature reduction to handle large volumes of data. The ultimate goal is to classify tracks into ten distinct genres while carefully managing the complexity of the dataset.

2 Related Works

The study of spectral features in audio signals has been extensively developed, given its usefulness in a wide range of applications. For instance, such features are widely adopted in speech recognition [1] and in sound recognition for hazardous situations [2]. A more specific yet less explored application is the classification of music tracks by genre, which represents the focus of this project.

A central concept in these studies is the use of Mel Frequency Cepstral Coefficients (MFCCs). MFCCs represent the short-term power spectrum of a sound, modeled according to the human auditory system's perception of frequency. In practice, they capture timbral and spectral characteristics that are crucial for distinguishing between different audio sources. Alongside MFCCs, other features are often considered, such as spectral contrast, chroma features, tempo, zero-crossing rate (ZCR), root mean square (RMS) energy, and spectral bandwidth.

In [3], the authors propose several feature extraction methods for audio classification, including MFCCs, and evaluate their performance using classifiers such as Grid Search (GS), Gaussian Neural Networks (GNN), and k-Nearest Neighbors (K-NN). Their study highlights the effectiveness of combining multiple features with different machine learning approaches.

In contrast, [4] introduces a feature extraction method based on Wavelet Coefficient Histograms (WCHs), which enables audio signals to be analyzed at multiple resolutions. This approach aims to better capture spectral characteristics by exploiting both time and frequency domain information, offering a complementary perspective to traditional MFCC-based techniques.

3 Dataset

I decided to use the GTZAN dataset of audio tracks, which contains 100 .wav files of about 30 seconds for each of the 10 musical genres, that are: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, rock, for a total of 1000 files. For this project I preferred to split those files into 3 seconds tracks, this way I have more data to train the Machine Learning models, obtaining almost 10.000 tracks, which are not perfectly balanced because of some shorter tracks in the original dataset. I chose not to use the dataset with pre-extracted features in order to make my analysis even more different from [3], starting from the initially extracted features.

To extract spectral and musical features, I relied on the Librosa library, which provides intuitive tools for audio analysis. By iterating the analysis over 3-second segments of each track, I obtained the following features: the mean and standard deviation of the first 15 MFCCs, the mean of the first 12 chroma values, the mean of the first 7 spectral contrast values, and the mean of the spectral centroid, bandwidth, rolloff, ZCR, RMS, and tempo, for a total of 55 features, excluding the "genre" feature, which is the target class and it's extracted from the name of the file. This preliminary feature extraction phase reduced the input size from 1.3 GB to approximately 6 MB.

3.1 Data Exploration

Firstly, I decided to check the balancing of the target class "genre", which is, as easily predictable from the dataset generation, well balanced. Some classes have less than 1000 samples, but the difference is minimal so I don't need any resampling technique.

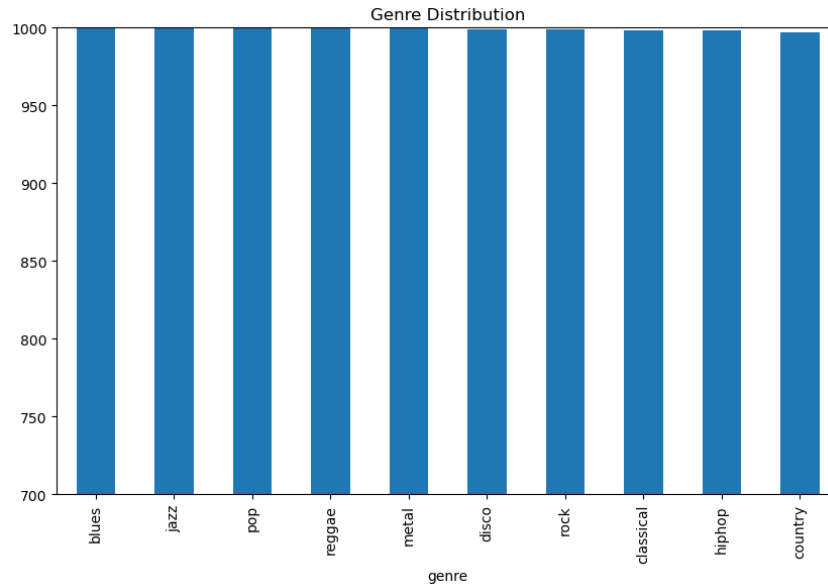


Figure 1: Genre Distribution

My analysis proceeded with the research of some feature values that could significantly help in the classification of the tracks. For example, I plotted the boxplots of "tempo" and "zcr_mean" for each class, obtaining different results as shown in the next figures.

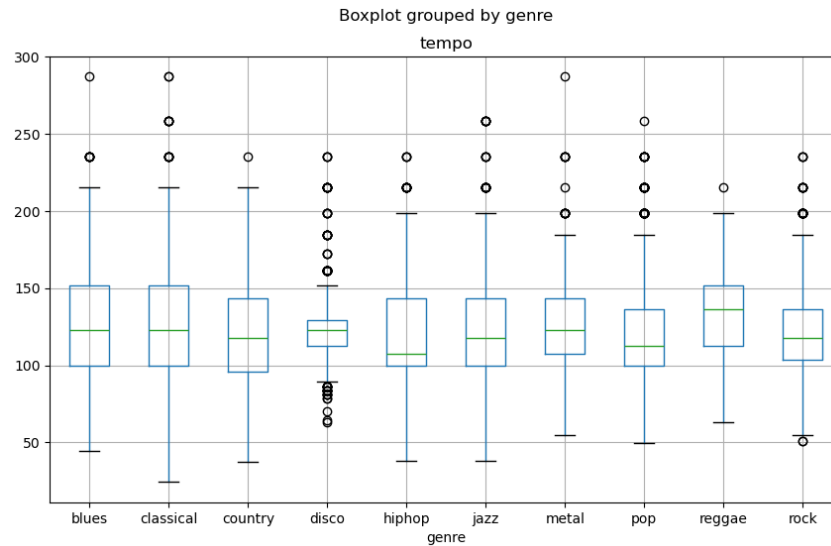


Figure 2: Boxplots on the feature "tempo"

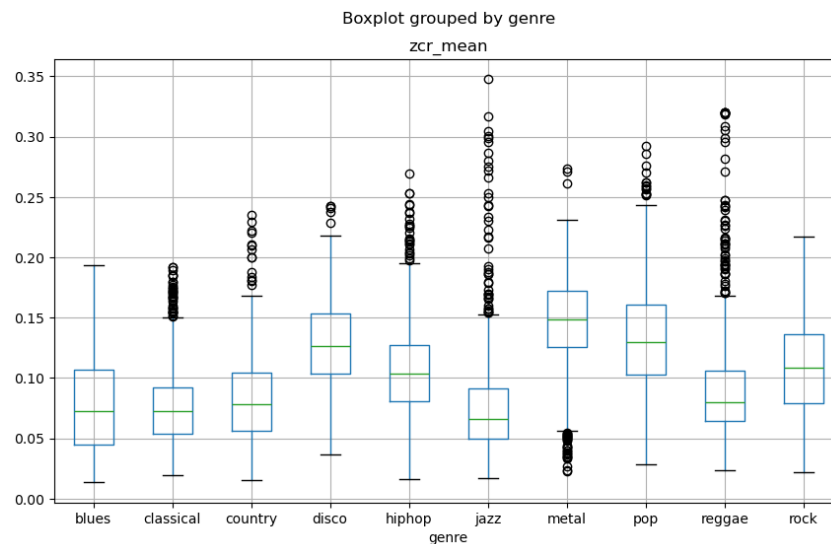


Figure 3: Boxplots on the feature "zcr_mean"

Continuing the analysis, I created scatterplots using some pairs of features as samples, in order to obtain a graphical representation of the differences between the various music genres. The pairs are:

- tempo – zcr_mean
- mfcc_1_mean – mfcc_2_mean
- spec_centroid_mean – spec_rolloff_mean
- chroma_1_mean – contrast_1_mean

As expected, it is not possible to perform a classification based solely on the observation of these plots, and this, like the rest of the data exploration I carried out, justifies the use

of machine learning models. However, it is interesting to note that centroid and rolloff appear to be somewhat correlated.

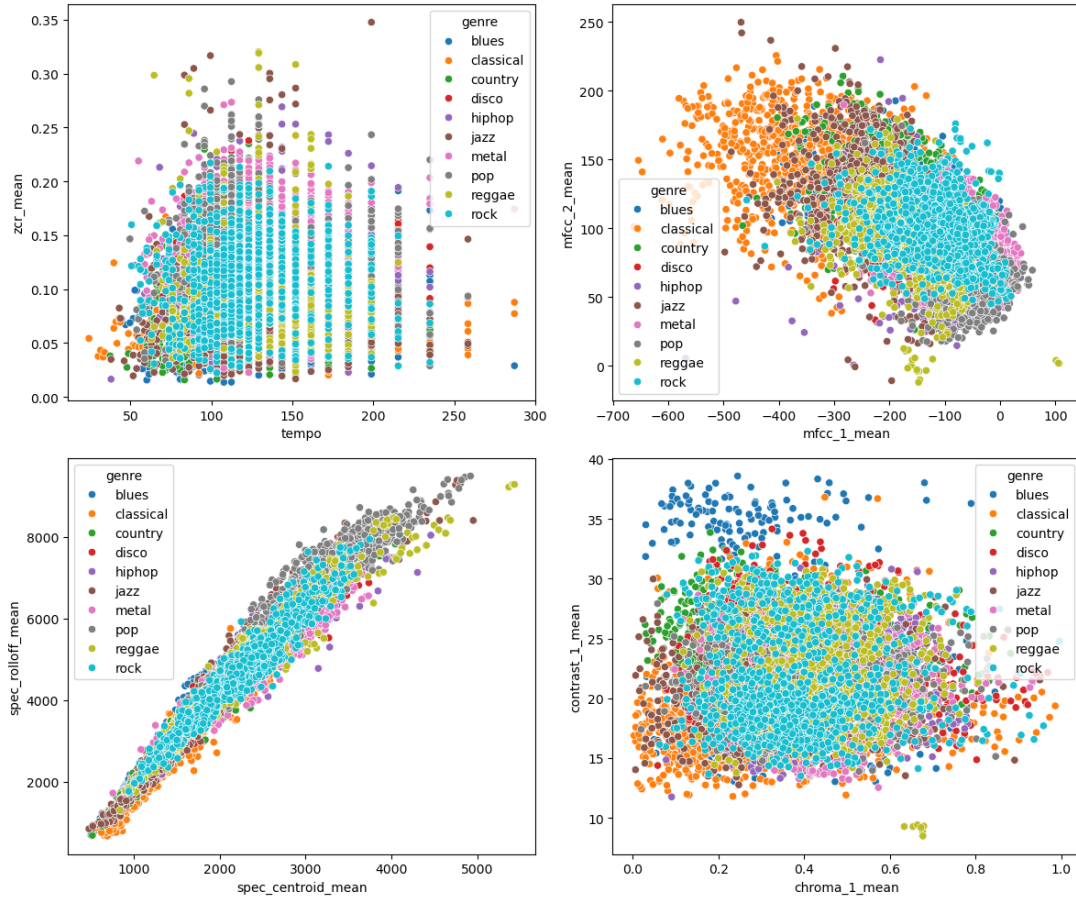


Figure 4: Scatterplots on different pairs of features

I also tried discretizing the features tempo and zcr_mean using both width-based and frequency-based approaches. This way, I could observe whether certain musical genres were easily identifiable because they were predominantly concentrated within specific value ranges. Regarding tempo, only Disco shows a strong presence in one of the frequency-based bins, although not enough to significantly aid classification. As for the zcr_mean analysis, no particular class stands out more than the others.

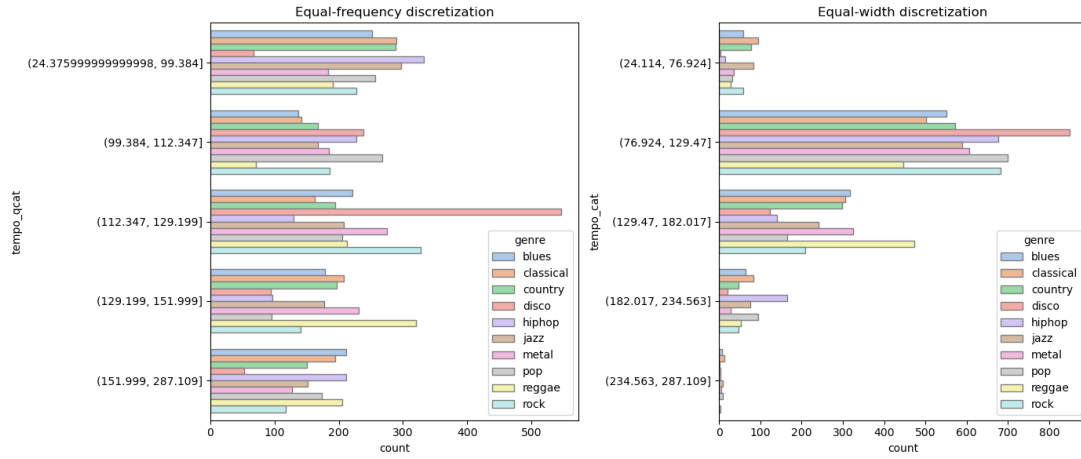


Figure 5: Discretization on the feature "tempo"

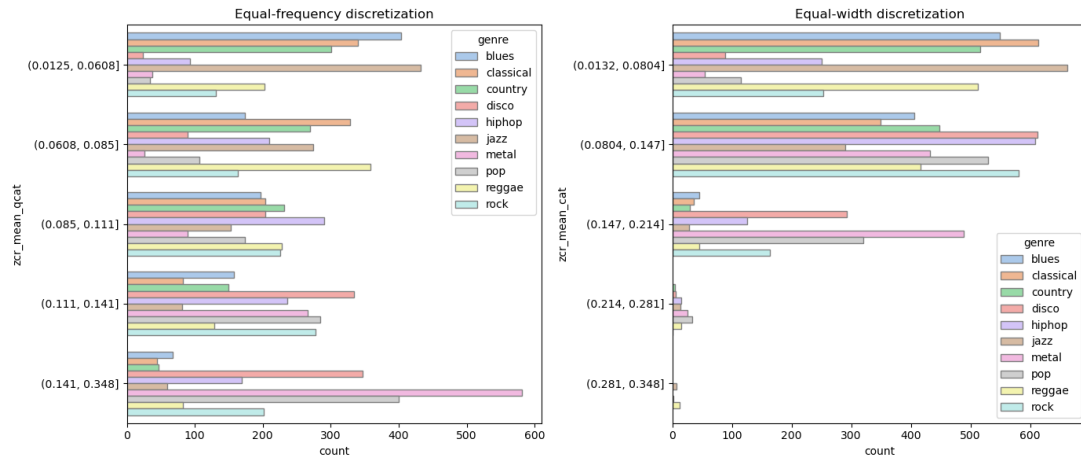


Figure 6: Discretization on the feature "zcr_mean"

The preliminary analysis of the dataset concludes with the generation of correlation matrices. For clarity, the analysis was divided into "mean" and "std" variables. The correlation patterns reveal that certain pairs of features are strongly correlated in the first matrix—for instance, specific contrast values show high correlation between each other, as well as spectral centroid, bandwidth, rolloff, and zcr. In contrast, features such as mfcc, chroma, and rms exhibit low correlation with the rest. These observations could support an initial feature reduction by removing highly redundant variables. However, in this case, only a few features would be eliminated, so I chose to retain all features and instead apply alternative reduction methods, which are discussed in the following sections.

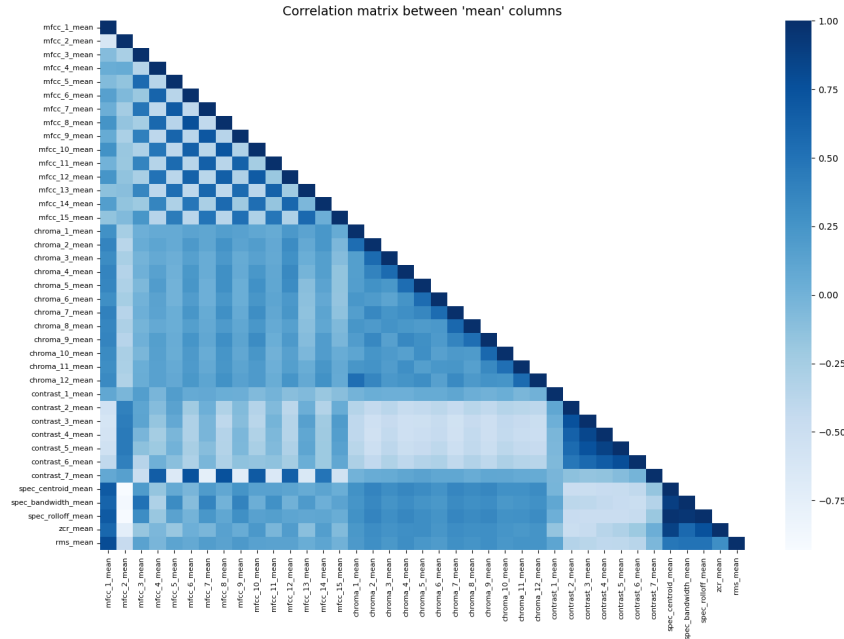


Figure 7: Correlation Matrix of the "mean" features

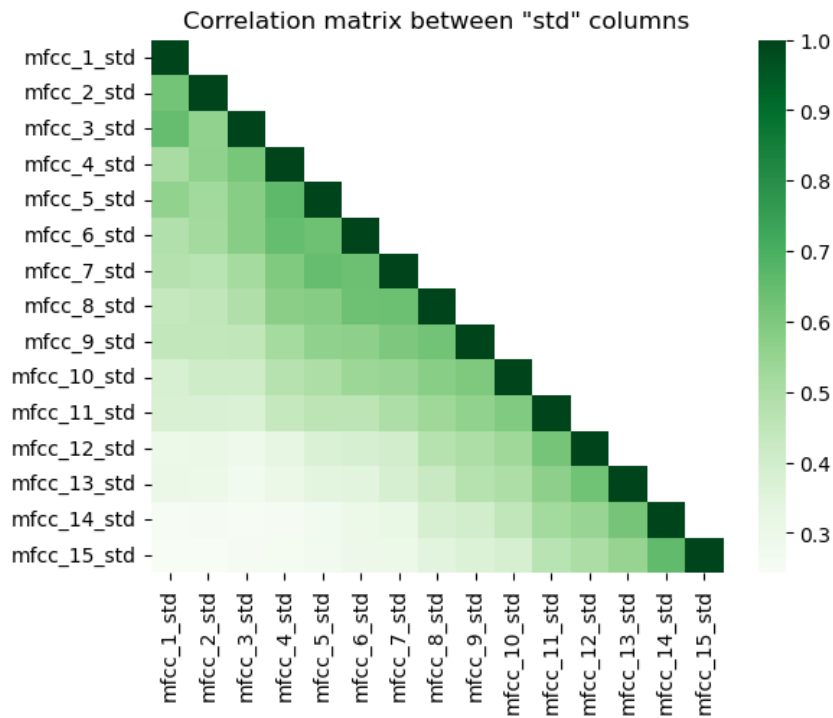


Figure 8: Correlation Matrix of the "std" features

3.2 Data Preprocessing

The dataset consists entirely of numerical values, with the only exception being the target variable, genre. Moreover, it does not present any issues related to class imbalance or missing values. Therefore, the only preprocessing step performed was the application of Scikit-learn's StandardScaler to obtain a scaled version of the data, which helps improve

the performance of the classification models. The scaler was fitted only after splitting the dataset into training and test sets, in order to prevent data leakage.

The genre column, although composed of strings, does not require explicit encoding because all the models tested are from Scikit-learn, which internally uses a LabelEncoder to convert string labels into numerical values suitable for classification.

3.3 Data Reduction

A primary objective of this project was to classify the musical genre of tracks while maintaining a balance between classification accuracy and the dimensionality of the input data. To address this, I experimented with multiple feature reduction techniques in parallel, aiming to identify the approach that would be most effective for the machine learning models evaluated during the model selection phase.

The models were tested on three types of input:

- Full dataset: 55 features
- PCA reduced dataset with 95% cumulative variance: 37 features
- RFE reduced dataset: 21-55 features (depending on the model)

4 Model Selection

In the Model Selection phase, four classifiers were trained with a GridSearch on different combinations of parameters, and the same training method was performed on the three input types discussed previously.

Model	Parameter	Values
Logistic Regression	C	{0.1, 1, 10}
	penalty	{l2}
	solver	{lbfgs}
SVM (LinearSVC)	C	{0.1, 1, 10}
Decision Tree	max_depth	{None, 5, 10, 20}
	min_samples_split	{2, 5, 10}
	min_samples_leaf	{1, 2, 4}
Random Forest	n_estimators	{100, 500}
	max_depth	{None, 10, 20}
	min_samples_split	{2, 5}

Table 1: Model Selection parameters

This selection of parameters aims to find the best fitting model while being lightweight and not computationally expensive to focus more on the next model optimization phase that will be heavier. The best combination of parameters for each model, also called the "best model", slightly differs between different inputs, and it's summarized in the next table:

Model	Parameter	Full	PCA	RFE
Logistic Regression	C	10	10	10
	penalty	l2	l2	l2
	solver	lbfgs	lbfgs	lbfgs
SVM (LinearSVC)	C	10	10	1
Decision Tree	max_depth	20	10	20
	min_samples_split	2	2	2
	min_samples_leaf	1	1	1
Random Forest	n_estimators	500	500	500
	max_depth	20	20	None
	min_samples_split	2	2	2

Table 2: Best Models for each Input

Considering that the training dataset is balanced, even if not perfectly, I chose to compare those best models using the accuracy metric, and the results are shown in the next heatmap:

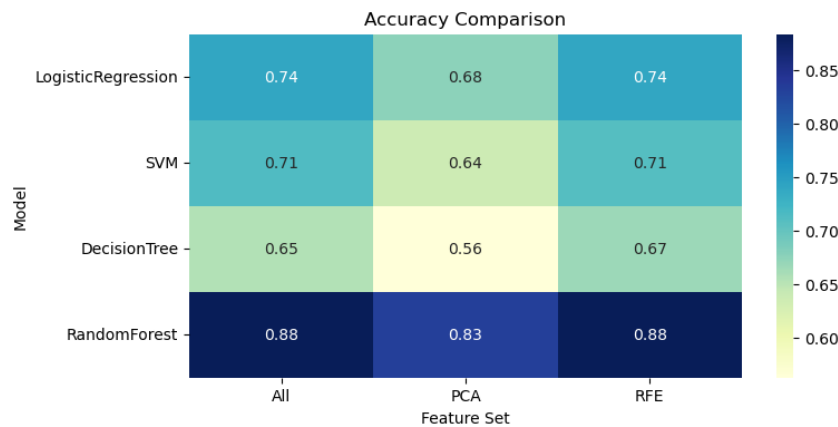


Figure 9: Accuracy heatmap for model selection

Considering these results, it's obvious that the RandomForest Classifier is the model that fits the best for this project. I decided to proceed with the optimization of the RandomForest model, using the feature set obtained through Recursive Feature Elimination, which selected 30 features with this model. This decision is based on the observation that the reduced feature set achieved an accuracy comparable to the full dataset, while substantially decreasing the dimensionality, which is particularly beneficial in terms of computational efficiency for large-scale applications.

5 RandomForest Classifier

5.1 Performance Optimization

In order to achieve the best performance possible in terms of classification quality, several combinations of parameters have been tested in the training phase of the model, resulting in an extensive GridSearch with Cross-Validation. The studied parameters and the resulting best model are the following:

Parameter	Values	Best Model
n_estimators	{100, 500, 1000}	1000
max_depth	{None, 20, 50}	20
min_samples_split	{2, 10, 20}	2
min_samples_leaf	{1, 5, 10}	1
max_features	{sqrt, log2, None}	sqrt
bootstrap	{True, False}	False

Table 3: Parameters grid considered for RandomForest.

In the next figure the main metrics for the performance evaluation of the best model are plotted for each target class, while the overall accuracy is 90%:

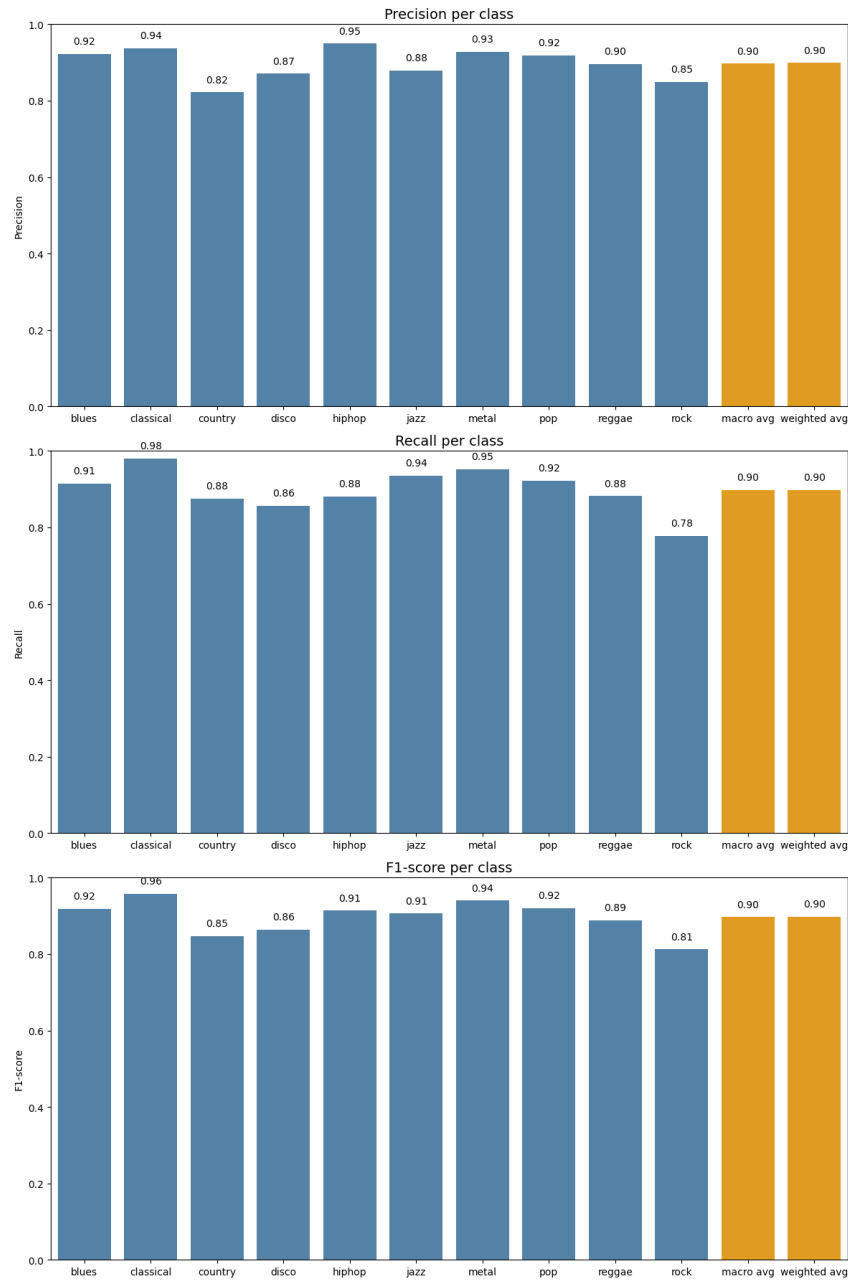


Figure 10: Precision, Recall and F1-score of the best RF model

The confusion matrix shown in the next figure indicates that, as expected, most samples are classified correctly. However, some classes are slightly favored by the model, such as country, disco, and jazz, while other classes, like rock and hip-hop, are more frequently confused with others. The classes that proved to be the easiest to distinguish are classical and metal.

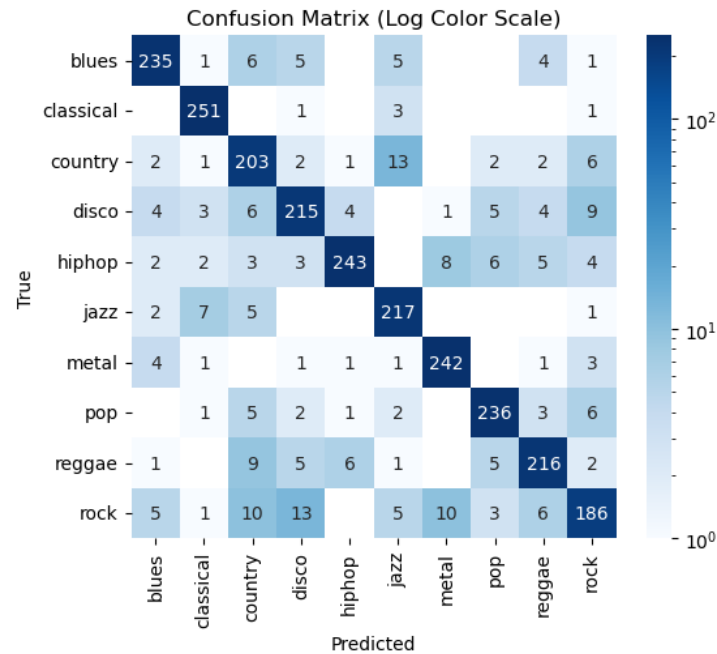


Figure 11: Best RF model confusion matrix

Then, I decided to plot the ROC curve of the best RF model, which in a multiclass setting represents the model's ability to distinguish each class from the others, showing the trade-off between the true positive rate (TPR) and the false positive rate (FPR) for each class. The results can be considered satisfying, with an average AUC around 0.99. The only two classes showing lower performance are hip-hop and rock, because their samples are sometimes confused with those of other similar classes, leading to more false positives and therefore a lower TPR in the ROC function.

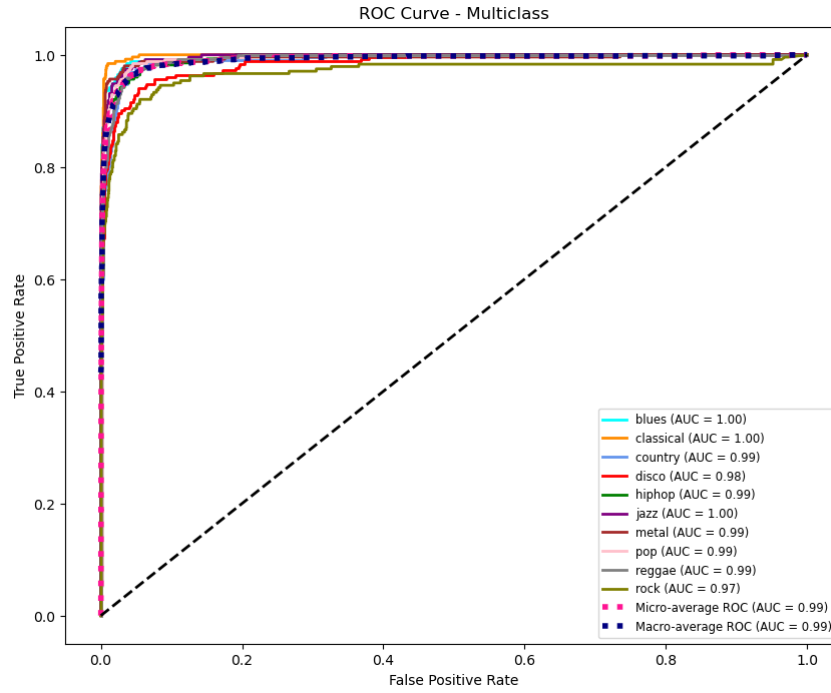


Figure 12: Receiver Operating Characteristic Curve

5.2 Model Explainability

In order to explain the decisions made by the RandomForest model, the feature importance histogram can provide a visual representation of which are the most considered features. In particular, "rms_mean" and "contrast_1" are the most significant attributes, as shown in the next figure.

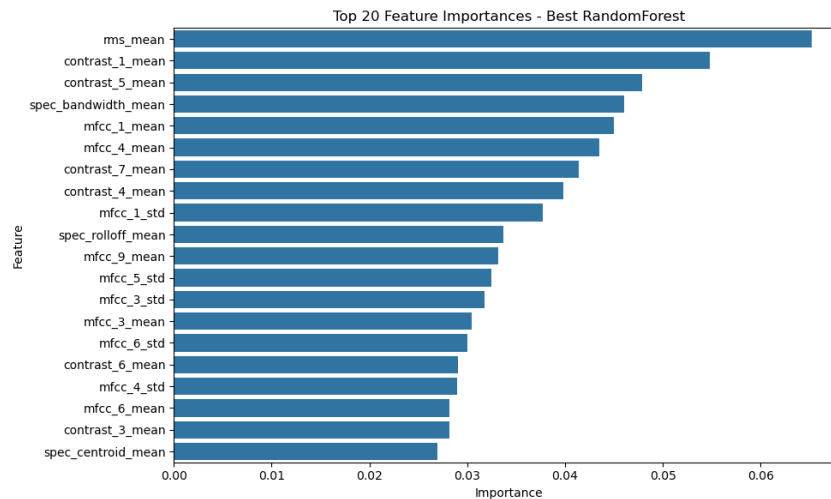


Figure 13: Feature Importance of the best model

To better understand the influence of different values of the features on the final decision of the model, I focused my analysis on the two classes which represent the easiest and the hardest genre to distinguish, classical and rock. On these two classes I plotted the SHAP summary plots, which visually show the most important features for the prediction of the class, and which values of those features were crucial.

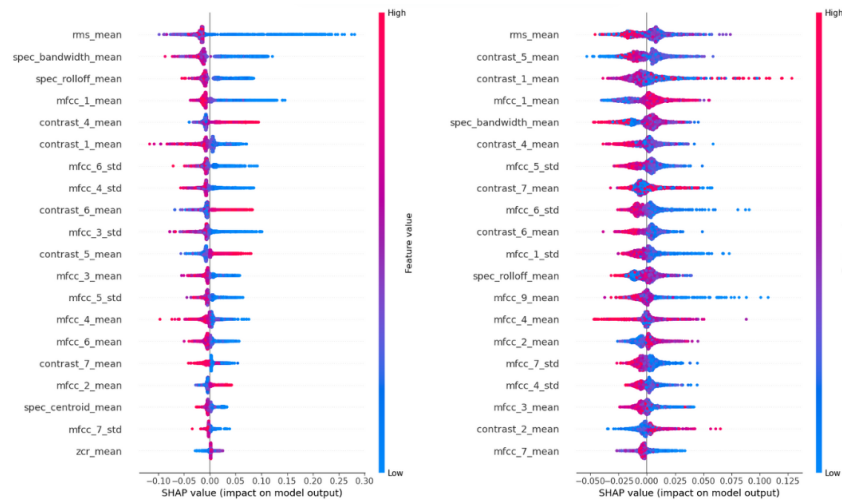


Figure 14: SHAP values for "classical" and "rock" classification

The optimization phase proved useful in improving the performance of the Random Forest model identified during model selection, even though the improvements were relatively small. This suggests that exploring more hyperparameter combinations in the Grid Search could lead to even better results. The best model is summarized in Table 3 and will be used to classify audio provided by users in the GUI, which will be described in the next chapter.

6 User Interface

For demonstration purposes, I decided to develop two simple GUIs. The first accepts a single audio track as input and outputs the model's prediction along with all classification probabilities, allowing a better understanding of the model's decision and whether the choice is confident or if other classes are almost equally likely. The second GUI has a more practical role: it takes multiple audio files as input and returns a list of files along with their predicted classifications, which is useful in real-world scenarios where large numbers of files need to be classified.

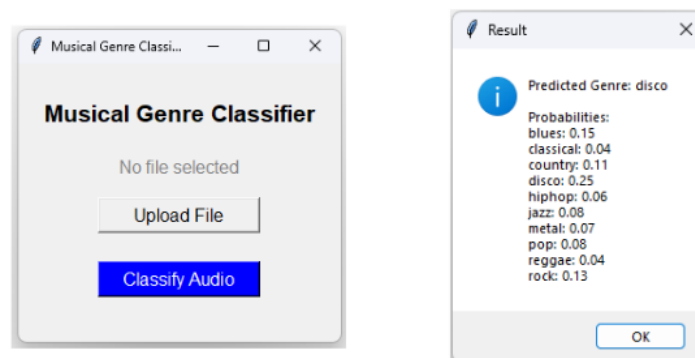


Figure 15: Classification probabilities GUI

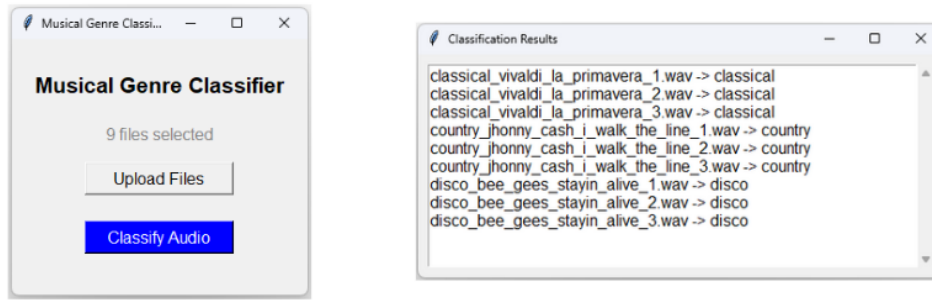


Figure 16: Multifile classification GUI

7 Conclusions

The development of this project has shown that common classification models can be used to automatically categorize audio tracks by musical genre, which is useful for applications such as streaming platforms handling large volumes of files. The combination of the Librosa library with a Random Forest model and RFE allowed for a significant reduction in input data size by extracting the most relevant features from audio tracks, enabling high-quality and easily understandable classification.

Future developments of this project should focus on expanding the initial dataset, as many tracks currently are too similar within the same genre, while each musical genre has many shades that can confuse even the best-performing model obtained. Additionally, it may be useful to increase both the number and range of parameters considered in the Grid Search for model optimization, in order to obtain Random Forest models that are increasingly more effective and well-suited for this application.

References

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